Lab - Research a Hardware Upgrade

Introduction

Use the Internet, a newspaper, or a local store to gather information about hardware components. Your customer's computer currently has one module of 2 GB of RAM, a 500 GB hard disk drive, and a PCIe video adapter card with 256 MB of RAM. Your customer wants to be able to play advanced video games.

Instructions

Step 1: Research memory options.

Shop around, and in the table below list the brand, model number, features, and cost for two different 8 GB modules of DDR3.

Brand and Model Number	Features	Cost
Kingston KVR1333D3N9/8G	*240-PIN UDIMM, NON-ECC, CL9,1.5V*	*₱1,250*
Ramsta Desktop LONG-DIMM DDR3 RAM 8GB 1600Mhz PC3-12800	*240-PIN UIDMM, NON-ECC,1.5V, PC3-12800*	*₱1,500*

Question:

Based on your research, which RAM would you select? Be prepared to discuss your decisions regarding the RAM you select.

I would go for the KINGSTON, since they are the same 8GB and it would be the cheaper option.

Step 1: Research hard disk drive options.

Shop around, and in the table below list the brand, model number, features, and cost for a 3 TB 7200 rpm SATA 3 hard disk drive and a 500GB or 1TB solid state drive (SSD).

Brand and Model Number	Features	Cost
Seagate Barracuda ST3000DM008	*3TB storage capacity, 7200 RPM spindle speed, SATA 3 (6Gb/s) interface, 256MB cache, 3.5-inch desktop form factor*	*₱4,586*
Samsung 990 PRO 1 TB NVMe/PCIe M.2 internal SSD MZ-V9P1T0BW	*1TB storage capacity, PCIe Gen 4.0 x4 NVMe interface, M.2 2280 form factor, up to 7,450 MB/s read speed, up to 6,900 MB/s write speed, Samsung V-NAND TLC, 1GB DRAM cache, AES 256-bit hardware encryption*	*₱15,995*

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Question:

Based on your research, which hard disk drive would you select? Be prepared to discuss your decisions regarding the hard disk drive you select.

I would select the Samsung SSD, since SSD are much faster than HDD. If possible I would use my SSD as my main ROM and add the Seagate Barracuda HDD as my extra ROM for storing files.

Step 2: Research video adapter card options.

Shop around, and in the table below list the brand, model number, features, and cost for two different PCIe video adapter cards with at least 1 GB of RAM.

Brand and Model Number	Features	Cost
EVGA GeForce 210 1GB DDR3 (01G-P3-1313-KR)	*1GB DDR3 video memory, PCI Express 2.0 x16 interface, 64-bit memory bus, HDMI, DVI-I, and VGA outputs, low-profile (half-height) bracket, and supports small form factor (SFF) desktop PCs*	*₱737*
NVIDIA GeForce GT 210 1GB DDR2 64-bit PCIe 2.0 Graphics Card	*1GB DDR2 video memory, PCI Express 2.0 x16 interface, 64-bit memory bus, DVI, VGA, low-profile compatible, and suitable for basic display, office work, and multimedia playback*	*₱719*

Question:

Based on your research, which video adapter card would you select? Be prepared to discuss your decisions regarding the video adapter card you select.

I would select the EVGA GeForce, it is already in DDR3 much faster than DDR2 and it offers HDMI as it is usually the main connector of the PC unit to the monitor.

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